

TSA6213 SYSTEM ADMINISTRATION & MAINTENANCE

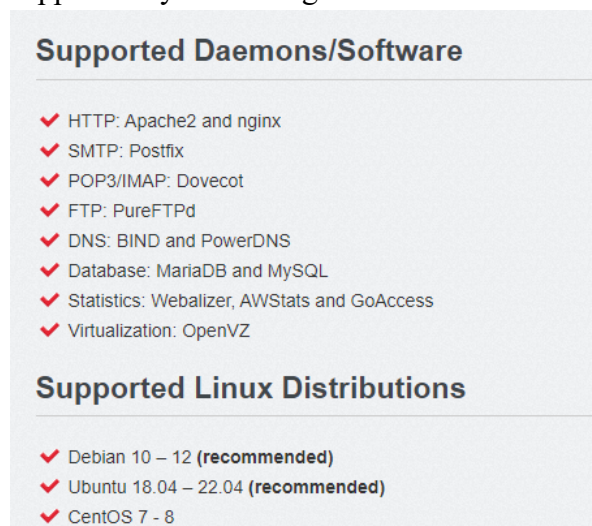
LAB 6

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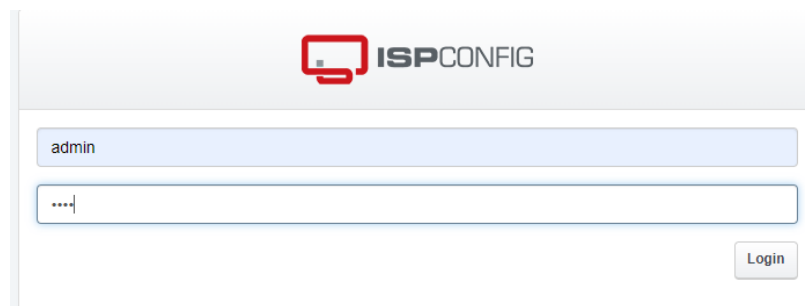
Internet Ready Environment - ISPConfig

1. ISPConfig is a free open-source control panel application that allows website owners to easily administer their sites, similar to cPanel and Plesk. It also allows resellers to manage multiple accounts on multiple physical or virtual servers. Go to ISPConfig website at <https://www.ispconfig.org/> .

a. Based on the latest information currently available, identify latest daemon and Linux distro supported by ISPConfig.

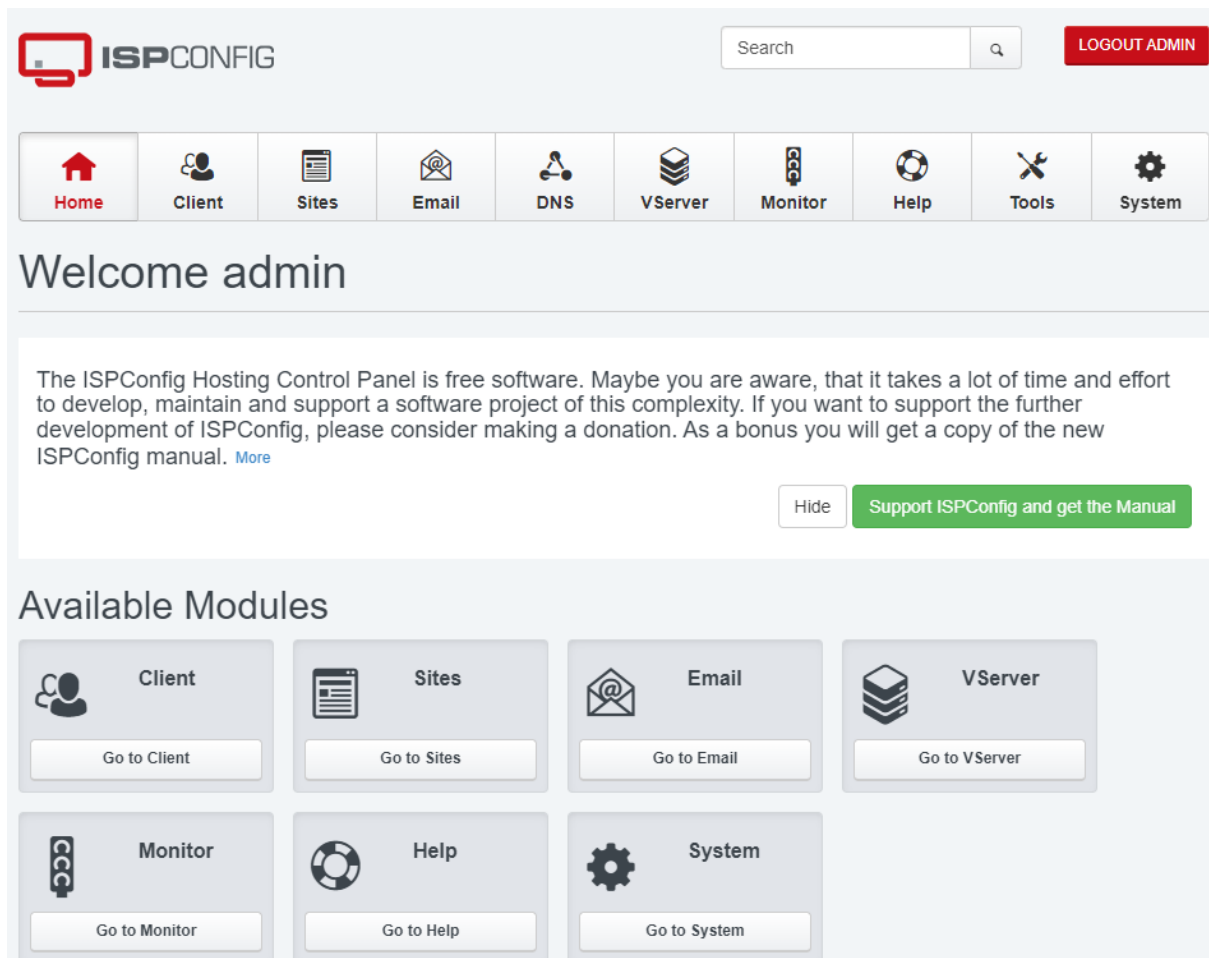


b. Click the online demo and login as Administrator. Identify how many modules supported under Admin.



Go to <http://demo3.ispconfig.org/>

username: admin | password: demo



The screenshot shows the ISPConfig Admin interface. At the top, there is a header with the ISPConfig logo, a search bar, and a 'LOGOUT ADMIN' button. Below the header is a navigation bar with icons and labels for Home, Client, Sites, Email, DNS, VServer, Monitor, Help, Tools, and System. The main content area starts with a 'Welcome admin' message. Below this is a paragraph of text about the software being free and a request for donations, with a 'More' link. To the right of this text are 'Hide' and 'Support ISPConfig and get the Manual' buttons. Further down is a section titled 'Available Modules' which contains seven cards: Client, Sites, Email, VServer, Monitor, Help, and System. Each card has an icon, a title, and a 'Go to [Module Name]' button.

ISPCONFIG

Search

LOGOUT ADMIN

Home Client Sites Email DNS VServer Monitor Help Tools System

Welcome admin

The ISPConfig Hosting Control Panel is free software. Maybe you are aware, that it takes a lot of time and effort to develop, maintain and support a software project of this complexity. If you want to support the further development of ISPConfig, please consider making a donation. As a bonus you will get a copy of the new ISPConfig manual. [More](#)

Hide Support ISPConfig and get the Manual

Available Modules

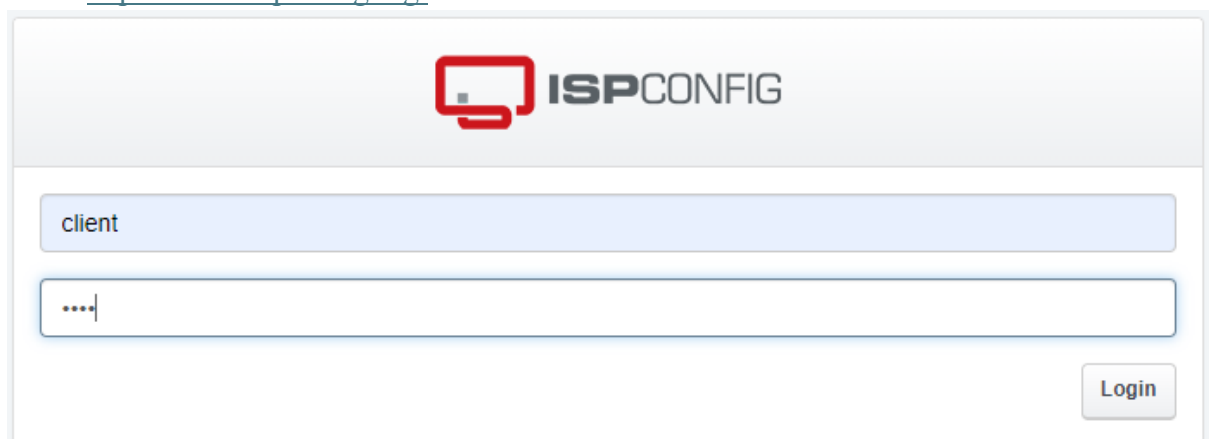
Client	Sites	Email	VServer
Go to Client	Go to Sites	Go to Email	Go to VServer

Monitor	Help	System
Go to Monitor	Go to Help	Go to System

There are 7 available modules supported under admin with 2 additional configurable modules which is DNS and Tools.

c. Click the online demo and login as Client. Identify how many modules supported under Client.

Go to <http://demo3.ispconfig.org/>



The screenshot shows the ISPConfig Client login interface. It features the ISPConfig logo at the top. Below the logo is a login form with two input fields: the first is labeled 'client' and the second is a password field with a masked input (dots). A 'Login' button is located to the right of the password field.

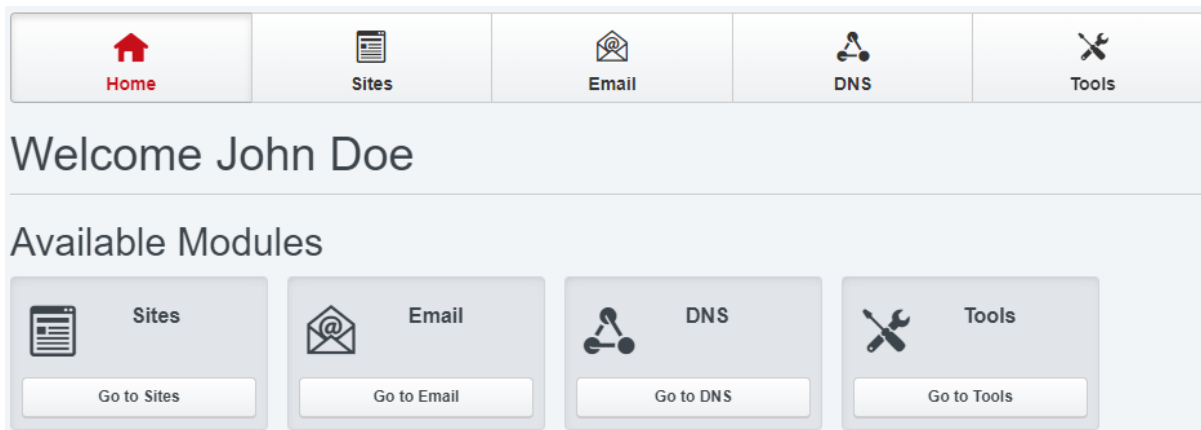
ISPCONFIG

client

...

Login

username: client | password: demo



There are 4 available modules supported under client.

Internet Ready Environment - Login Remotely (SSH)

2. Install SSH Server, use `sudo apt-get install openssh-server` and confirm the installation by logging-in using local machine. At terminal, use `ssh username@localhost` , if your username is osboxes, give `ssh osboxes@localhost` and give password for osboxes.

> `sudo apt-get install openssh-server`

```
osboxes@osboxes:~$ sudo apt-get install openssh-server
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libssh2-1 mc-data
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  ncurses-term openssh-client openssh-sftp-server ssh-import-id
Suggested packages:
  keychain libpam-ssh monkeysphere ssh-askpass molly-guard
The following NEW packages will be installed:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
The following packages will be upgraded:
  openssh-client
1 upgraded, 4 newly installed, 0 to remove and 141 not upgraded.
1 not fully installed or removed.
Need to get 1,359 kB of archives.
After this operation, 6,027 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

> `sudo systemctl status ssh` [to check status of ssh]

```

osboxes@osboxes:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: en
   Active: active (running) since Thu 2024-12-19 02:41:21 UTC; 4min 45s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Main PID: 18067 (sshd)
      Tasks: 1 (limit: 4592)
     Memory: 1.3M
    CGroup: /system.slice/ssh.service
            └─18067 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups

Dec 19 02:41:21 osboxes systemd[1]: Starting OpenBSD Secure Shell server...
Dec 19 02:41:21 osboxes sshd[18067]: Server listening on 0.0.0.0 port 22.
Dec 19 02:41:21 osboxes systemd[1]: Started OpenBSD Secure Shell server.
Dec 19 02:41:21 osboxes sshd[18067]: Server listening on :: port 22.

[1]+  Stopped                  sudo systemctl status ssh

```

> ssh osboxes@localhost

```

osboxes@osboxes:~$ ssh osboxes@localhost
The authenticity of host 'localhost (127.0.0.1)' can't be established.
ECDSA key fingerprint is SHA256:4uH74nVcOG8ouqJmOE7OfSEi7TvTKIj0DrKLra6/wd8.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added 'localhost' (ECDSA) to the list of known hosts.
osboxes@localhost's password:
Welcome to Ubuntu 20.04.4 LTS (GNU/Linux 5.4.0-200-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Thu 19 Dec 2024 02:46:41 AM UTC

System load:            0.05
Usage of /home:         0.2% of 249.18GB
Memory usage:           22%
Swap usage:             0%
Processes:              219
Users logged in:        1
IPv4 address for enp0s3: 192.168.63.207
IPv6 address for enp0s3: 2001:d08:1206:8aa9:a00:27ff:fe9a:17ef

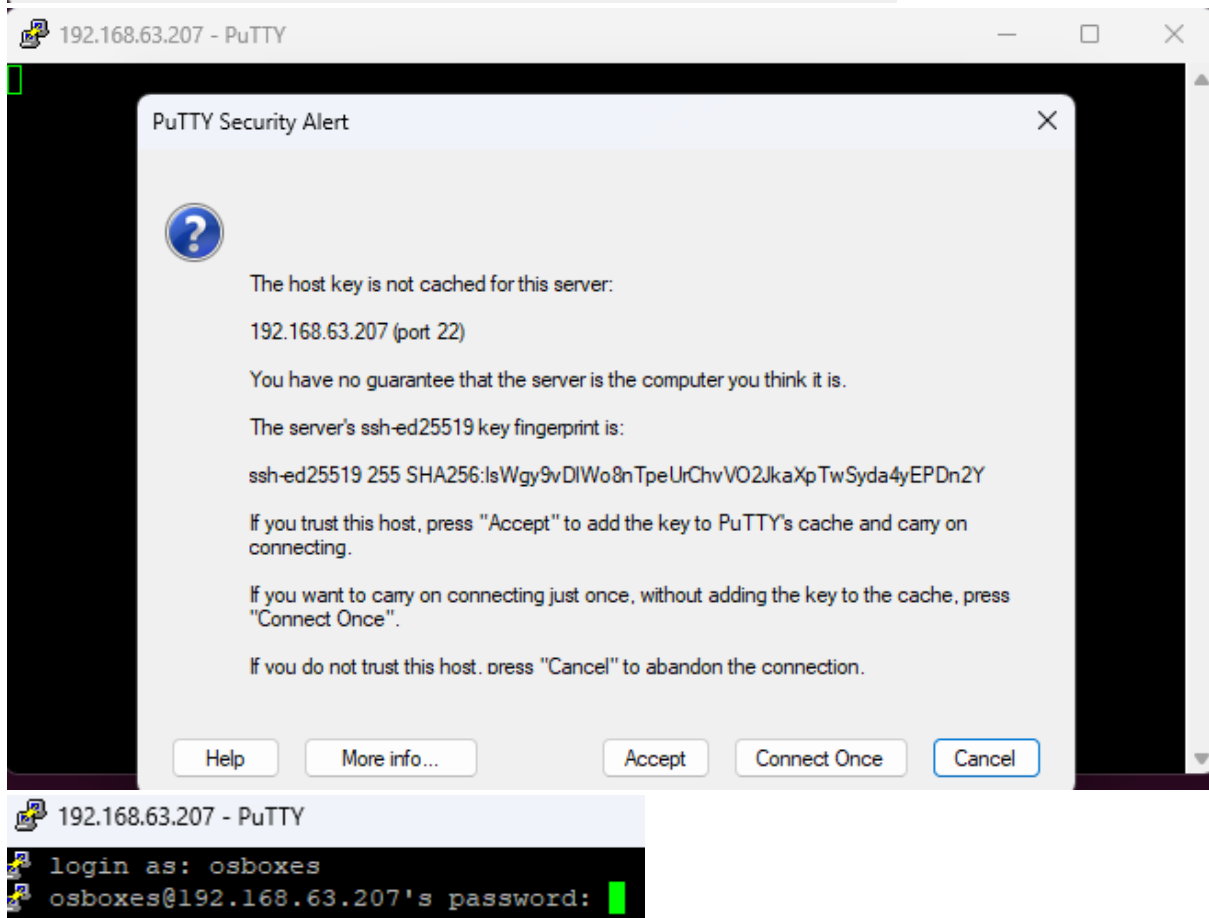
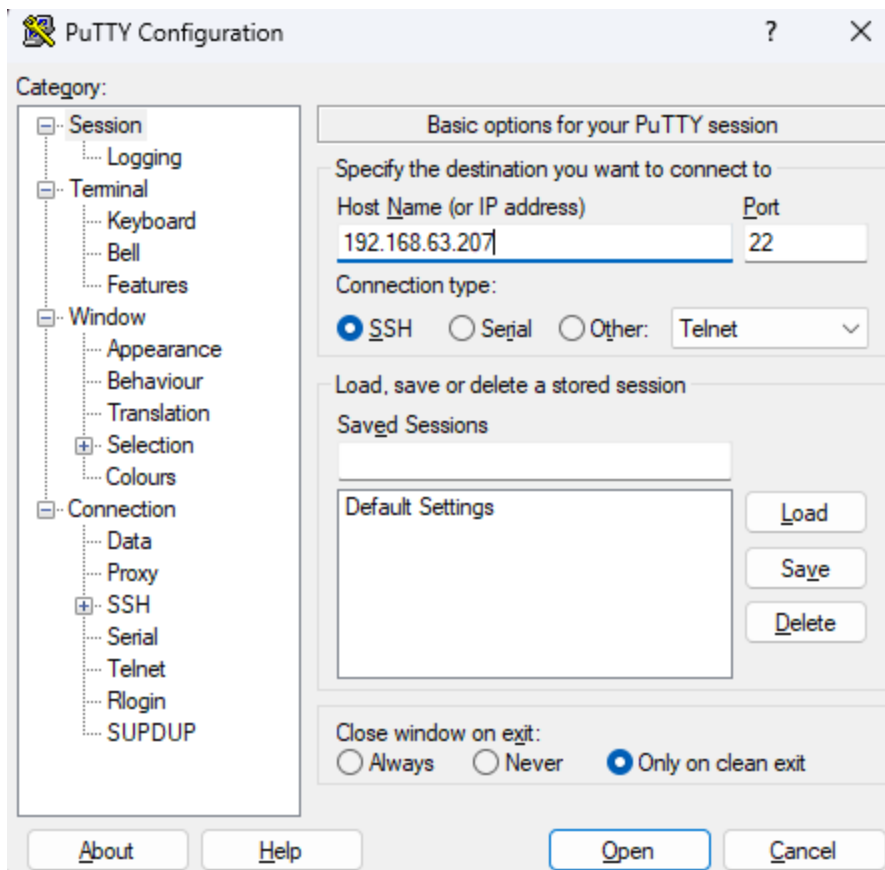
Expanded Security Maintenance for Applications is not enabled.

148 updates can be applied immediately.
101 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

2 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

```

3. Set your Virtual Machine network to **Host Only Adapter** and get the IP address. From your Windows machine login to SSH Server using Windows SSH Client (putty.exe).



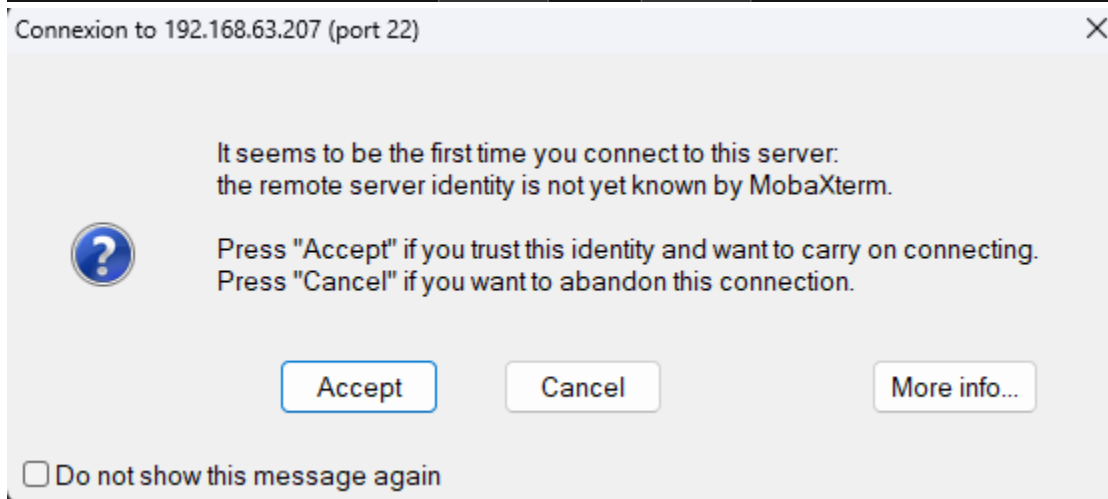
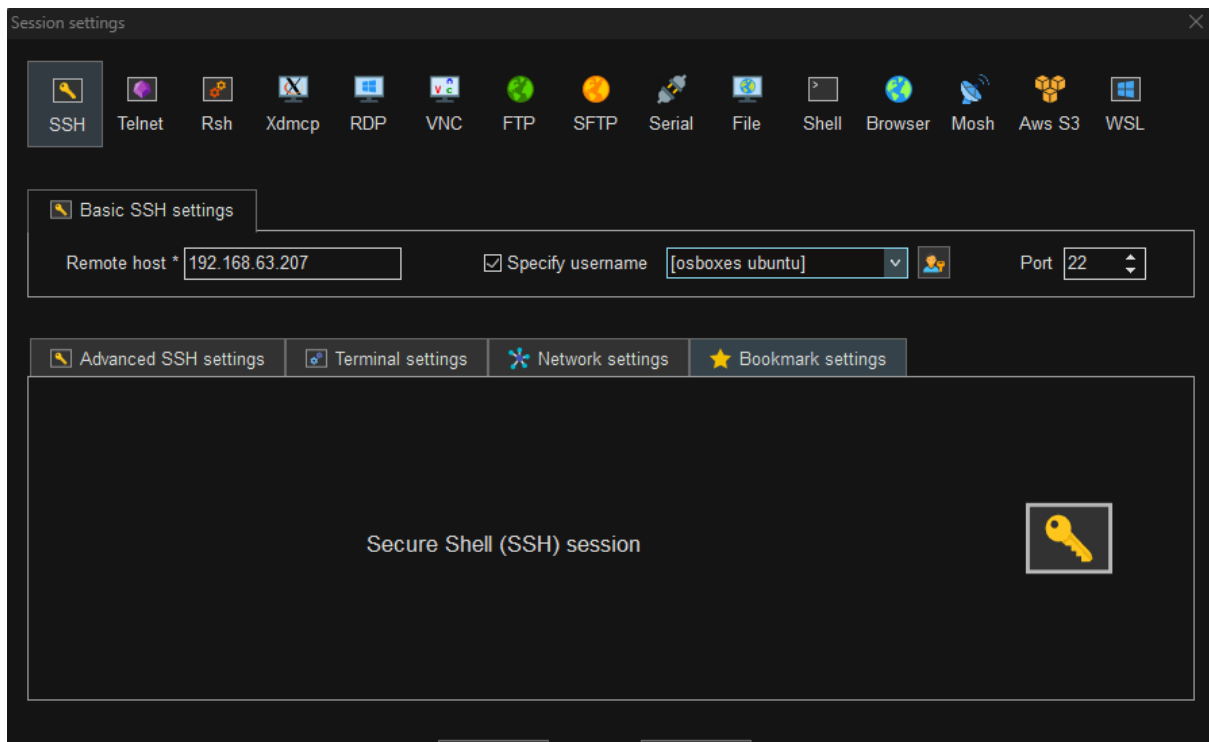
```
osboxes@osboxes: ~  
Welcome to Ubuntu 20.04.4 LTS (GNU/Linux 5.4.0-200-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/advantage  
  
System information as of Thu 19 Dec 2024 02:53:40 AM UTC  
  
System load:          0.01  
Usage of /home:       0.2% of 249.18GB  
Memory usage:        23%  
Swap usage:          0%  
Processes:           220  
Users logged in:      1  
IPv4 address for enp0s3: 192.168.63.207  
IPv6 address for enp0s3: 2001:d08:1206:8aa9:a00:27ff:fe9a:17ef  
  
Expanded Security Maintenance for Applications is not enabled.  
  
148 updates can be applied immediately.  
101 of these updates are standard security updates.  
To see these additional updates run: apt list --upgradable  
  
2 additional security updates can be applied with ESM Apps.  
Learn more about enabling ESM Apps service at https://ubuntu.com/esm  
  
New release '22.04.5 LTS' available.  
Run 'do-release-upgrade' to upgrade to it.  
  
Last login: Thu Dec 19 02:46:42 2024 from 127.0.0.1  
osboxes@osboxes:~$  
  
Last login: Thu Dec 19 02:46:42 2024 from 127.0.0.1  
osboxes@osboxes:~$ dir  
car Desktop Downloads Pictures Templates  
carl Documents Music Public Videos  
osboxes@osboxes:~$
```

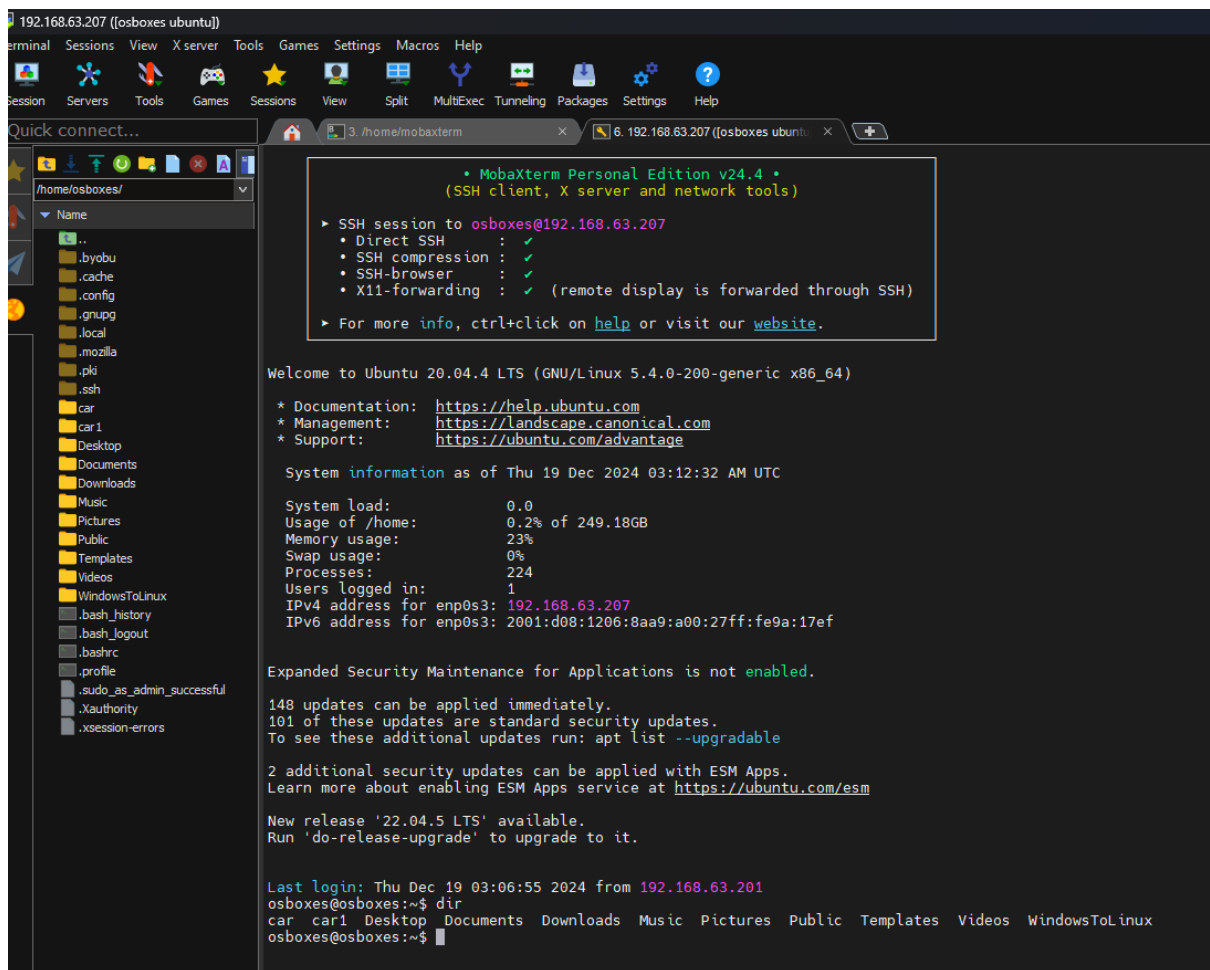
SSH: Secure Copy and Login using SSH Key

4. Secure Copy Protocol (SCP) is a means of securely transferring computer files between a local host and a remote host or between two remote hosts. It is based on the Secure Shell (SSH) protocol. "SCP" commonly refers to both the Secure Copy Protocol and the program itself. dirdir

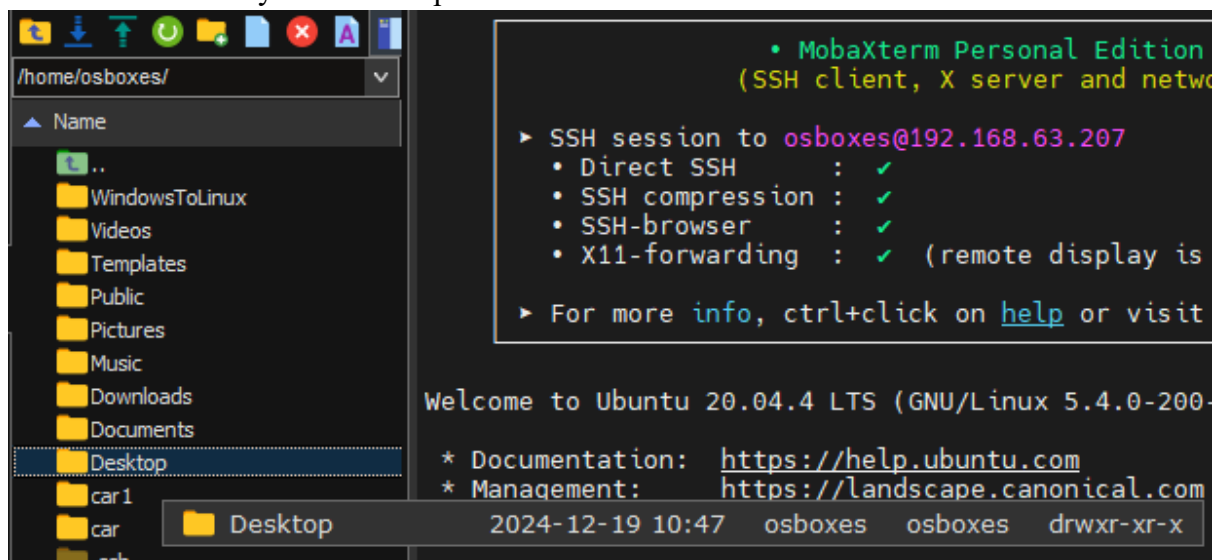
a. Use SCP program to copy files from your Windows PC to your Server. You can use MobaXTerm or pscp.exe as Client.

> open MobaXTerm

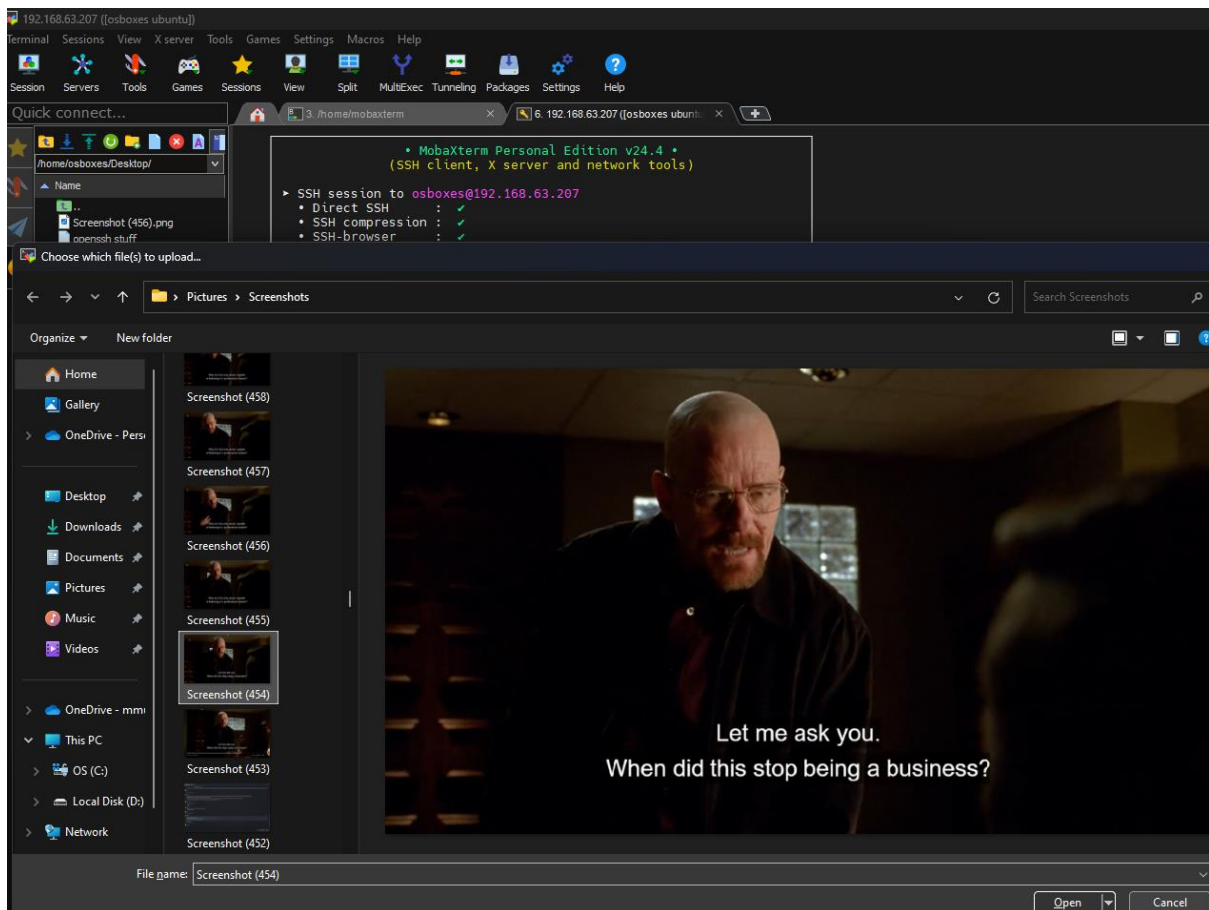




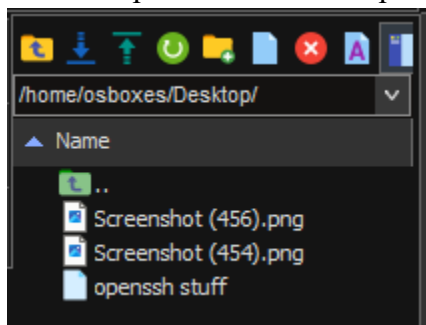
> Go to the directory we want to put stuff into.



> The third button from left, is the upload button.



- > Select the file we want to upload from our Windows PC to Linux.
- > Press Open and it will be uploaded to our Linux folder.

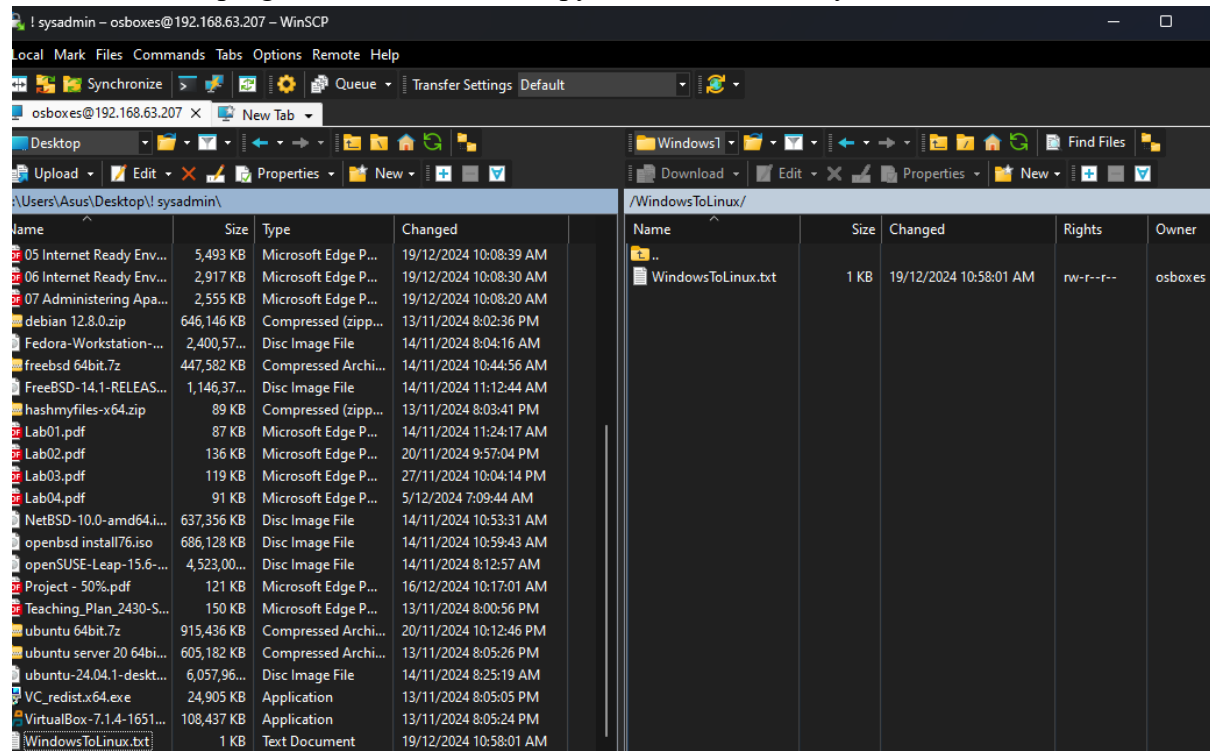


- > dir
- > cd Desktop
- > dir

```
osboxes@osboxes:~$ dir
car car1 Desktop Documents Downloads Music Pictures Public Templates Videos WindowsToLinux
osboxes@osboxes:~$ cd Desktop
osboxes@osboxes:~/Desktop$ dir
openssh\ stuff Screenshot\ (454).png Screenshot\ (456).png
osboxes@osboxes:~/Desktop$
```

- > Files are inside the directory.

b. Use WinSCP program in Windows to copy files and folder to your Linux Server.

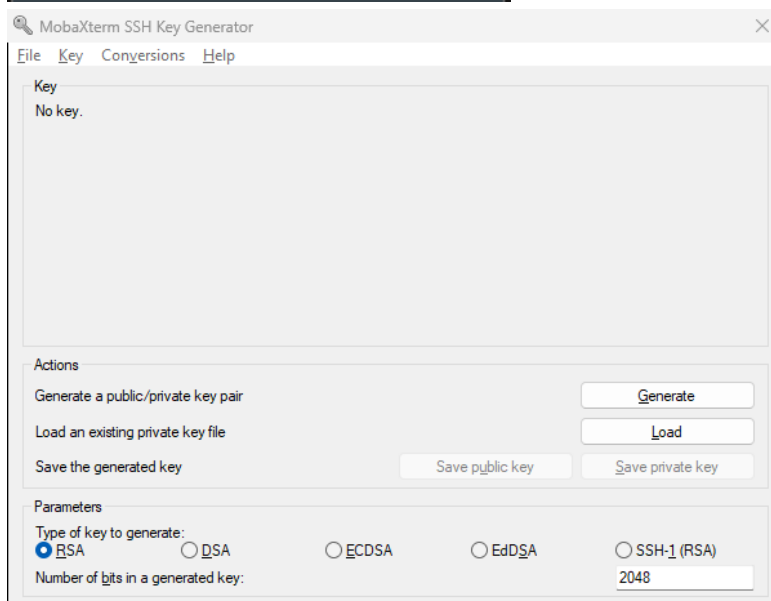
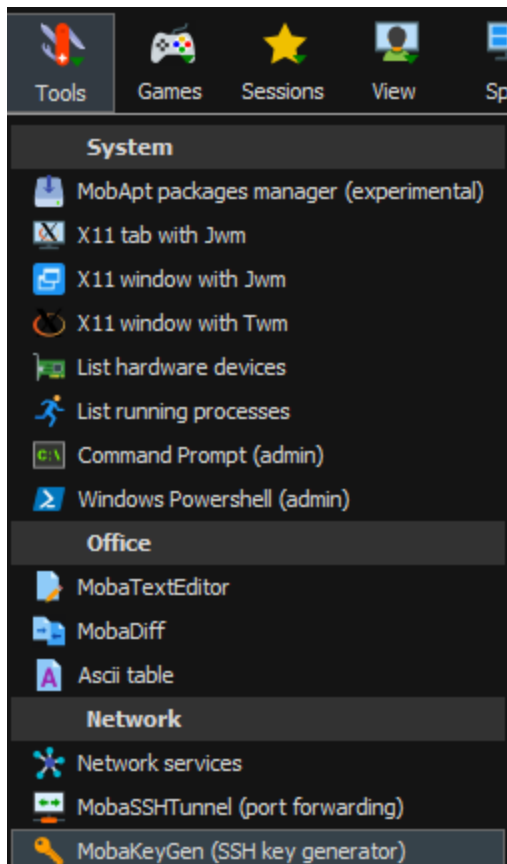


5. Secure Shell (better known as SSH) is a cryptographic network protocol which allows users to securely perform a number of network services over an unsecured network. SSH keys provide a more secure way of logging into a Server with SSH than using a password alone. While a password can eventually be cracked with a brute force attack, SSH keys are nearly impossible to decipher by brute force alone. Use MobaXterm to perform task as below:

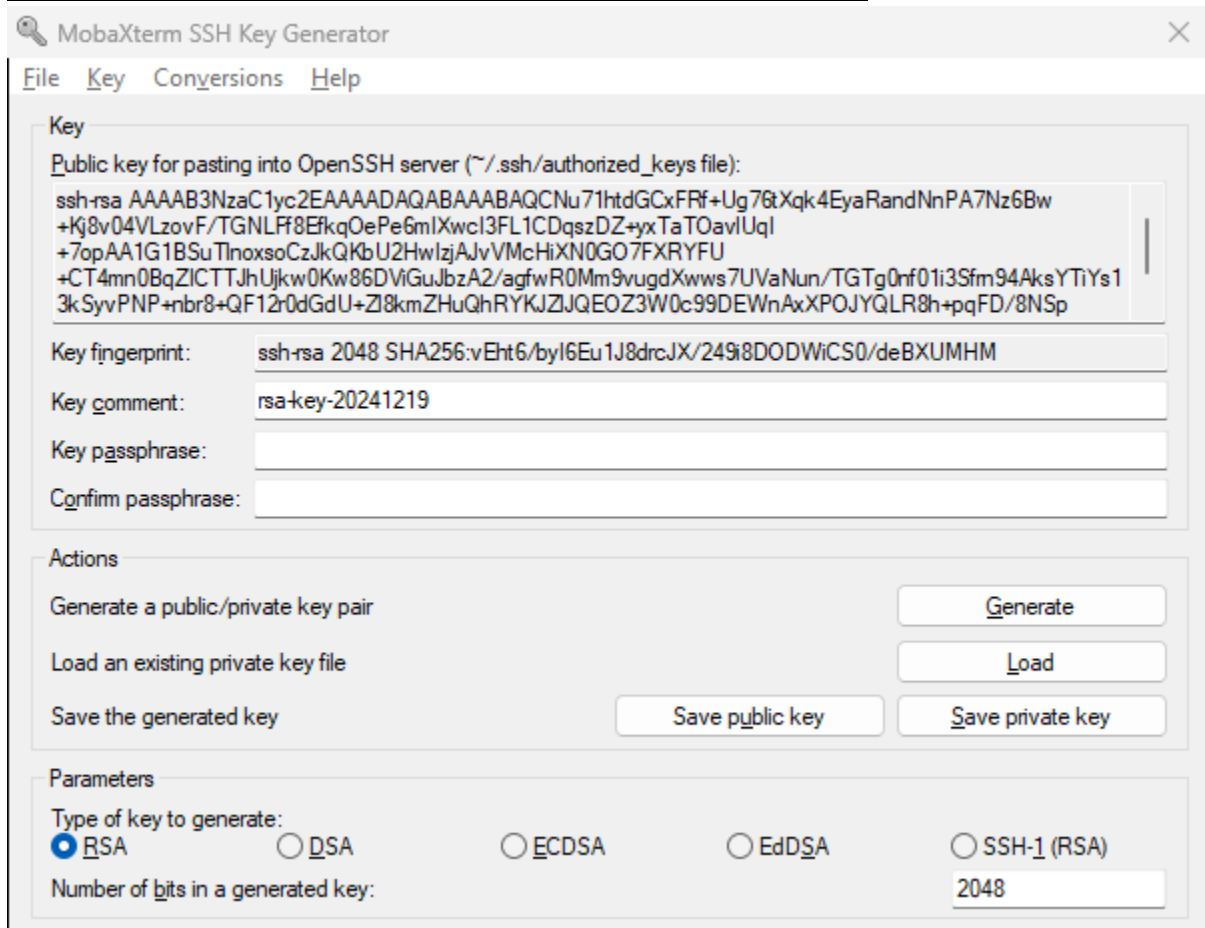
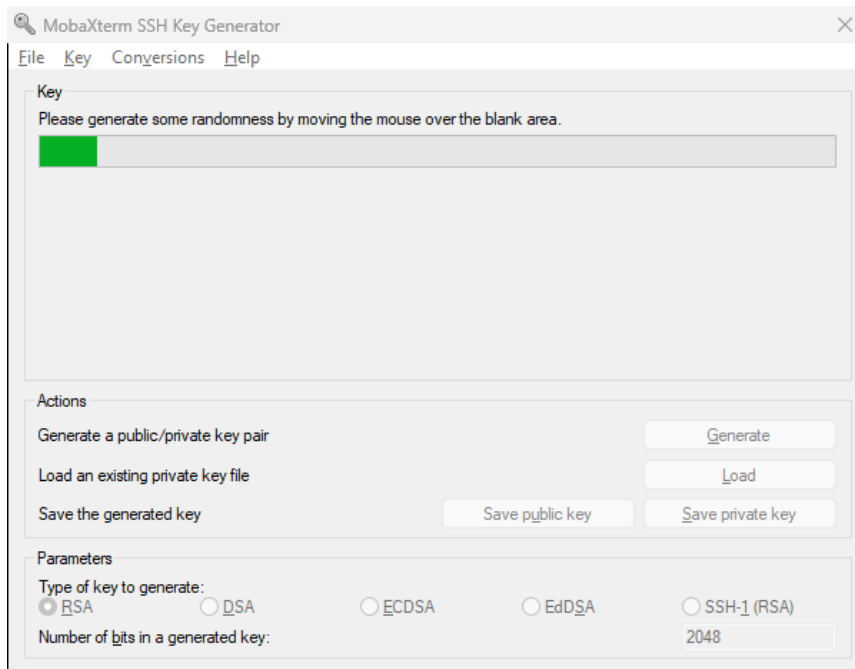
a. Generate key using RSA algorithm

> Tools

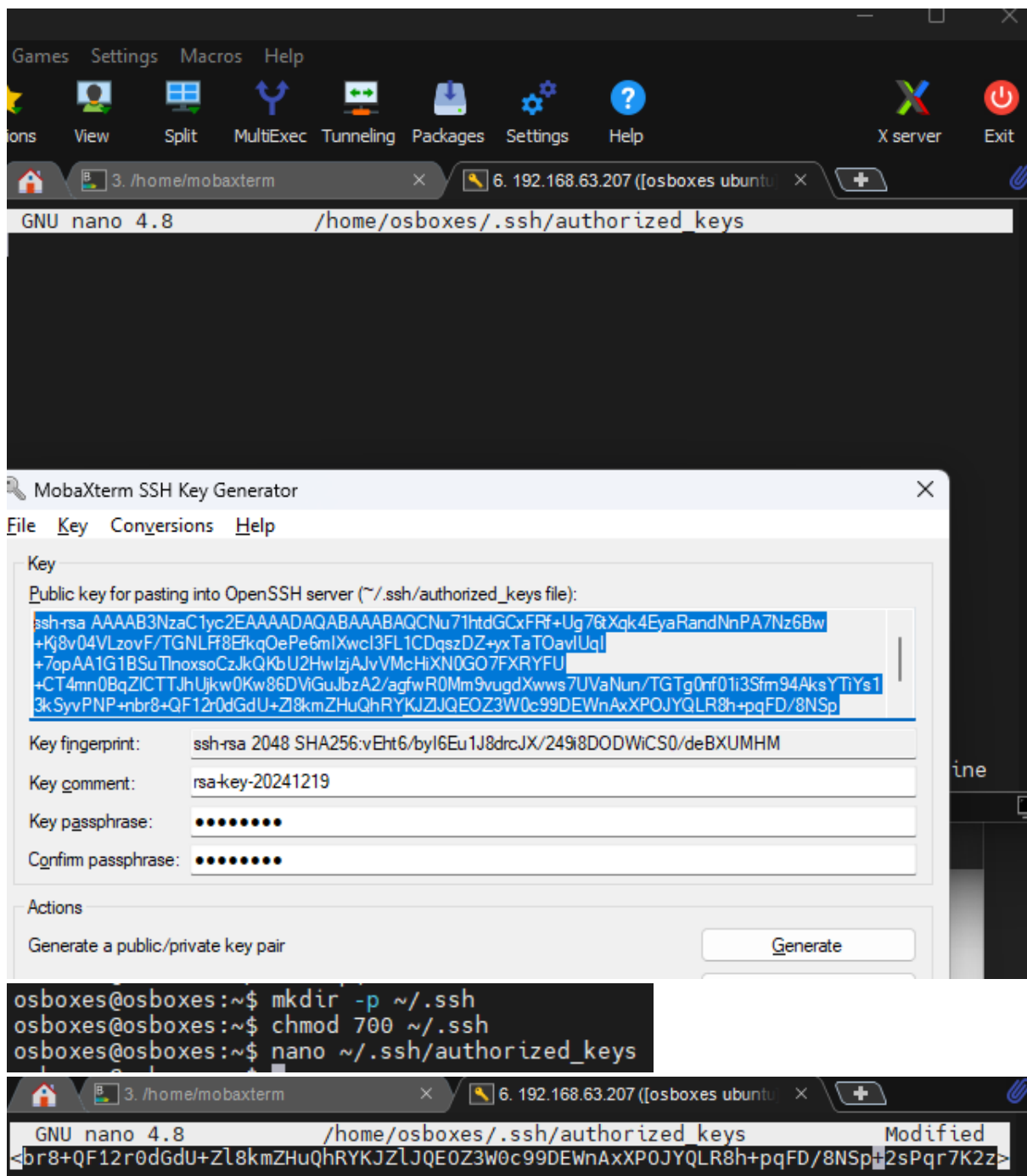
> MobaKeyGen (SSH key generator)



- > Select RSA
- > Press Generate



b. Copy the key from Client to Server



> Ctrl + O to save, Ctrl + X to exit.

```
osboxes@osboxes:~$ chmod 600 ~/.ssh/authorized_keys
```

c. Login to Server using SSH key

> Open Session, select SSH.

> Enter IP address at remote host, username, and use private key.

SSH

Telnet

Rsh

Xdmcp

RDP

VNC

FTP

SFTP

Serial

File

Shell

Browser

Mosh

Aws S3

WSL

Basic SSH settings

Remote host *

192.168.63.207

☒ Specify username

osboxes ubuntu

Port

22

Advanced SSH settings

Terminal settings

Network settings

Bookmark settings

☒ X11-Forwarding

☒ Compression

Remote environment:

Interactive shell

Execute command:

☐ Do not exit after command ends

SSH-browser type:

SFTP protocol

☐ Follow SSH path (experimental)

☒ Use private key

C:\Users\Asus\Desktop\! sysadmi

Execute macro at session start: <none>

Expert SSH settings

7. 192.168.63.207 (osboxes ubuntu)

Authenticating with public key "rsa-key-20241219"

Passphrase for key "rsa-key-20241219":

• MobaXterm Personal Edition v24.4 •
(SSH client, X server and network tools)

► SSH session to osboxes@192.168.63.207

- Direct SSH : ✓
- SSH compression : ✓
- SSH-browser : ✓
- X11-forwarding : ✓ (remote display is forwarded through SSH)

► For more info, ctrl+click on help or visit our website.

Welcome to Ubuntu 20.04.4 LTS (GNU/Linux 5.4.0-200-generic x86_64)

* Documentation: <https://help.ubuntu.com>

* Management: <https://landscape.canonical.com>

* Support: <https://ubuntu.com/advantage>

System information as of Thu 19 Dec 2024 03:40:36 AM UTC

System load: 0.02

Swap usage: 0%
Processes: 228
Users logged in: 1
IPv4 address for enp0s3: 192.168.63.207
IPv6 address for enp0s3: 2001:d08:1206:8aa9:a00:27ff:fe9a:17ef

Expanded Security Maintenance for Applications is not enabled.

148 updates can be applied immediately.
101 of these updates are standard security updates.
To see these additional updates run: `apt list --upgradable`

2 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at <https://ubuntu.com/esm>

New release '22.04.5 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Dec 19 03:12:32 2024 from 192.168.63.201